UNIVERSITY LOGO

AVISHKAR 2026

Maharashtra State Inter-University Research Convention

Zonal Round

Topic: iTantra — Offline Neural Speech Transceiver for Low-Bitrate Radio Links

Team: YOLO **Category: Engineering & Technology** **Code: SIH26173** **Level: UG**

ABSTRACT

Voice is too data-heavy for a low-bitrate link, but in a distress scenario voice reaches everyone — literate or not. iTantra recognises speech on the phone, sends only the recognised text, and speaks it out on the other phone. On-device ASR runs offline in 10 Indian languages; a message travels as a 49–315-byte packet over Bluetooth or Wi-Fi Direct — around 65× smaller than the same speech as audio. A delivery layer — ACKed outbox, receipts, heartbeats, alert escalation, kill survival — keeps every message whole when the link drops. Mean character error rate **0.0918** over 2,000 clips across 10 languages; the two-phone loop has run on physical phones.

HYPOTHESIS

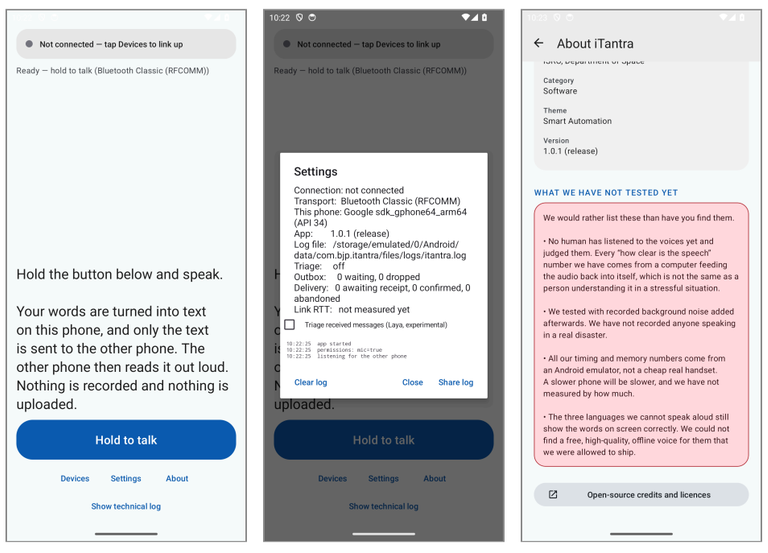
1. On-device speech recognition is accurate enough to replace the cloud in an emergency workflow.
2. Text is a viable wire format: a few hundred bytes per sentence survive a low-bitrate link that audio cannot.
3. Pruning the model's output vocabulary to the ten required scripts eliminates cross-script noise — the model cannot emit off-script characters.
4. Deterministic delivery — ACKed outbox, receipts, ordered resends — makes a lossy link behave predictably; alerts must preempt normal speech.
5. A killed app can resume pending messages from disk with no silent loss.
6. Low-RAM phones can host a 349 MB model if idle memory is returned under pressure.

FIXED FOR ALL BUILDS

Rationale (high level): only text crosses the link — ~7 s of speech ≈ 200 B, about 65× smaller than a voice call carrying the same speech. Reliability is treated as a delivery problem, not a radio problem: every segment is ACKed or resent (3/6/12 s → 15 s backoff, max 10 tries), 4 s heartbeats run both ways, a peer is declared gone at 12 s. One shipped configuration for every run: arm64-v8a, minSdk 23, a 1,002-token pruned vocabulary, and 96 unit tests on every build.

Run	Configuration	Test set	Result
A — baseline	ASR v1, 10,288-token vocabulary	FLEURS, clean, 2,000 clips	Mean CER 0.0992
B	ASR v2	FLEURS, clean, 2,000 clips	Mean CER 0.0934
C — shipped	ASR v3, pruned 10,288 → 1,002 tokens	FLEURS, clean, 2,000 clips	Mean CER **0.0918**; zero out-of-script characters
D	C, pink noise @ 15 dB SNR	Hindi clips	Script retention 40% → **78%**
E	C, noise @ 0 dB SNR	Tamil clips	44% retention — weakest case, declared
F — device	Shipped build, on-device self-tests	Android 34 emulator, arm64	3 self-tests green; ASR RTF 0.156; TTS RTF 0.078

EVIDENCE — THE REAL APP

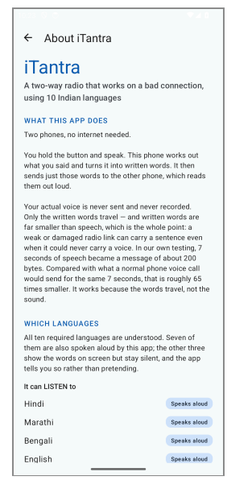


Release build v1.0.1 on Android 34 — home, settings, and the About screen that states the project's own limits. Two-phone loop demonstrated on physical phones, 27 Sep 2026.

INTRODUCTION

Problem statement SIH26173 asks for a TTS & STT aided neural transceiver radio for low-bitrate links (ISRO, Department of Space). In floods, disasters and remote posts, the radio link is weak and intermittent — exactly when a message must get through. Voice is the natural interface, but raw or even compressed voice overwhelms a damaged link and dies on packet loss; cloud speech services need internet, which is absent by definition.

Our approach inverts the stack: speech is recognised offline on the sender phone, and only text crosses the radio. Text is compact, orderable and acknowledgeable, so the link can drop and recover without losing the message; the receiver synthesises it aloud. No cloud, no accounts — the app does not even request the INTERNET permission.



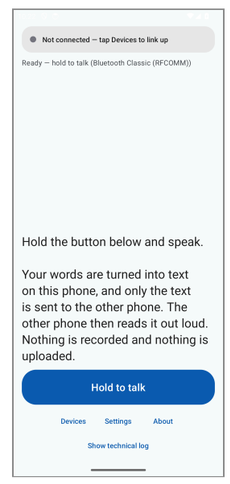
SYSTEM COMPONENTS

Component	Size / spec	Role
Offline ASR — Omnilingual character-level CTC, int8, vocabulary pruned to the ten scripts	349 MB	speech → text, 10 languages
VAD — Silero	0.63 MB	segmentation; end-of-speech flush
Neural TTS — Piper voices (hi, mr, bn, ta, te, ml, en)	≈60 MB per voice (one resident)	text → speech, 7 of 10 languages
Wire format — JSON envelope: id, lang, ttl, type	49–315 B per message	the radio payload
Transport — Bluetooth RFCOMM (primary), Wi-Fi Direct (fallback)	2 Mbit/s class	link layer
Delivery layer — ACKed outbox, receipts, heard-ids ledger, PendingStore	temp-file + rename JSON	order, dedupe, kill survival
Optional triage — Laya int8, off by default	143 MB	destructive-language messages → priority alert

App: Kotlin, arm64-v8a, minSdk 23 — signed release APK v1.0.1, 993 MB, zero network permissions.

RESULTS AND CONCLUSION

Shipped v1.0.1: a 993 MB signed APK, Android 6+, arm64, zero network permissions. Measured: mean CER **0.0918** across 2,000 FLEURS clips (10 languages, 6.7 h of real speech); **zero** out-of-script characters; Hindi script retention under pink noise 40% → **78%**; the script filter drops foreign-script segments whole, 25/25 cases; Hindi TTS round-trip CER 0.339 → **0.102**. On-device: ASR RTF 0.12–0.16, TTS RTF 0.055–0.08 → **≈1 s mouth-to-ear**; 428 MB PSS with the model resident. 96 unit tests and three on-device self-tests pass; the two-phone walkie-talkie loop ran end-to-end on two physical phones (27 Sep 2026). **Conclusion:** a text-first neural transceiver makes voice serviceable over low-bitrate, unreliable links — fully offline, on commodity phones.



FUTURE SCOPE

1. Human listening panel and field trials — the noise sets are synthetic; real field audio is the next measurement.
2. Real low-end phone numbers: latency, RAM and battery on a budget handset.
3. Neural voices for the remaining three languages (Gujarati, Kannada, Odia — text-only today); Tamil robustness at 0 dB.
4. Language identification, to separate Hindi/Marathi and Bengali/Assamese.
5. Foreground service for very long holds; smaller APK.

REFERENCE

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6. espeak-ng pronunciation rules. github.com/espeak-ng/espeak-ng
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8. Team YOLO. iTantra — release v1.0.1, DELIVERY_FINDINGS, VERIFICATION_STATUS. github.com/psb-001/Itantra (2026)

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